

GEOTECHNICAL LABORATORY REPORT

GeoTech Soil Analysis Platform — Automated Engine

REPORT INFORMATION

Report Title: Atterberg Soil Characterization Report — Data Generated: 5/23/2026, 1:07:58 AM
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1. EXECUTIVE SUMMARY

This formal report outlines Atterberg limit classifications and subgrade suitability calculations for project "kigali rp soil". A total of 1 soil samples were laboratory analyzed. The testing indicates an average Liquid Limit of 40.0% and an average Plasticity Index of 20.0%, categorizing the active stratum domain as cohesive subgrade materials requiring strict compaction control.

2. ANALYSIS STATISTICS

TOTAL SOIL SAMPLES

1

AVERAGE LIQUID LIMIT

40.0%

AVERAGE PLASTICITY INDEX

20.0%

3. ENGINEERING CONCLUSIONS

- Physical limits indicate moderate plasticity and medium swelling potential.
- Sand-to-clay boundaries match target structural subgrade parameters.
- Particle distributions represent cohesive clay particles with high water-holding values.

4. GEOTECHNICAL RECOMMENDATIONS

- Chemical Lime stabilization (4-6% by mass) is recommended for CL/CH clay boundaries to limit expansion indices.
- Compaction must target minimum 98% Proctor Dry Density under strict moisture criteria.
- Provide adequate site drainage to protect structural subgrade from water pooling.

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